

# ANNIKA HO

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## TECHNICAL SKILLS

**Software / Programming:** SOLIDWORKS (Certified Associate), Onshape, MATLAB, AutoCAD, Click PLC, Microsoft Office Suite, C, C#  
**Manufacturing / Tools:** FDM 3D Printing, General Shop Tools, Manual Mill/Lathe, Laser Cutting  
**Electronics:** Microcontrollers, Soldering, Multimeter, Oscilloscope, Function Generator

## EDUCATION

### University of British Columbia

Bachelor of Applied Science - Mechanical Engineering

Sep. 2021 - Apr. 2026

Vancouver, Canada

## TECHNICAL WORK EXPERIENCE

### Global Gourmet Foods

Jun. 2024 - Dec. 2024

Mechanical Engineering Co-op

- Used **Onshape** to design a modular-body conveyor system, producing engineering drawings for sheet metal and weldments
- Optimized warehouse layout using **AutoCAD**, increasing pallet racking spaces by 15%
- Created a scalable budgeting spreadsheet using **MS Excel** VBA macros that consolidated expenses across 10+ project divisions
- Ensured secure electrical connections during **PLC** wiring using crimps, heat shrink, and strain relief

### NuWave Research

May 2023 - Aug. 2023

Mechanical Engineering Co-op

- Applied **ASME Y14.5 GD&T** standards to 50+ SOLIDWORKS drawings for mechanical parts and machinery
- Used **SOLIDWORKS Composer** to create and maintain SOPs for fabrication and assembly
- Collaborated with coworkers to troubleshoot mechanical issues in industrial microwaves
- Researched and compared products, providing supervisors with concise analytical reports

## PROJECTS

### 6DOF Prosthetic Hand Capstone

Sep. 2025 - Apr. 2026

- Collaborated with a multidisciplinary engineering team to adapt a commercial robotic hand for prosthetic applications
- Led in-person consultations with prosthetics clinics to identify user needs and medical regulations
- Designed and 3D printed a prosthetic hand mount in Onshape for demonstration
- Performed hardware validation and troubleshooting in an electronics laboratory

## DESIGN TEAM EXPERIENCE

### UBC Rapid 3D Printing Design

Sep. 2023 - May 2025

Filament Recycling Subteam

- Developed a **MATLAB** script for filament shredder to calculate shear and bending stresses on shaft
- Designed and modelled filament shredding blades and hexagonal shaft in **SOLIDWORKS**
- Performed online and in-person client consultation for 3d modelling and printing
- Collaborated with WiE (Women in Engineering) UBC to hold a 3D modelling and printing workshop for 30+ high school students

## VOLUNTEER EXPERIENCE

### Tetris Director

Jul. 2022 - Apr. 2025

UBC Esports Association

- Founded and spearheaded the **North American Collegiate Tetris League**, consisting of 44 teams and 250+ players
- Oversaw 20+ volunteers to manage event funding, streaming, and staffing
- Fostered a local community for puzzle gaming enthusiasts through tournaments and weekly meetups
- Produced Collegiate Tetris League streams weekly on Twitch, reaching up to 100 concurrent viewers
- Regularly managed the UBCEA gaming lounge, troubleshooting PCs and responding to member complaints